

Given:

To implement Decision Tree and Random Forest classifiers for breast cancer diagnosis, compare their performance, analyze feature importance, and evaluate the models using cross-validation.

DATASET: [Click to Download Heart Disease Dataset](#)

Decisiontree and RandomForest.py

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
df = pd.read_csv(r"..\DataSets\data.csv")
if "Unnamed: 32" in df.columns:
    df = df.drop(columns=["Unnamed: 32"])
df = df.drop(columns=["id"])
df["diagnosis"] = df["diagnosis"].map({"M": 1, "B": 0})
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
dt_pred = dt.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, dt_pred))
plt.figure(figsize=(20,10))
plot_tree(
    dt,
    feature_names=X.columns,
    class_names=["Benign", "Malignant"],
    filled=True,
    fontsize=8
)
plt.show()
dt_depth = DecisionTreeClassifier(max_depth=3, random_state=42)
dt_depth.fit(X_train, y_train)
depth_pred = dt_depth.predict(X_test)
print("Decision Tree Accuracy (max_depth=3):",
      accuracy_score(y_test, depth_pred))
rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)
print("Random Forest Accuracy:",
```

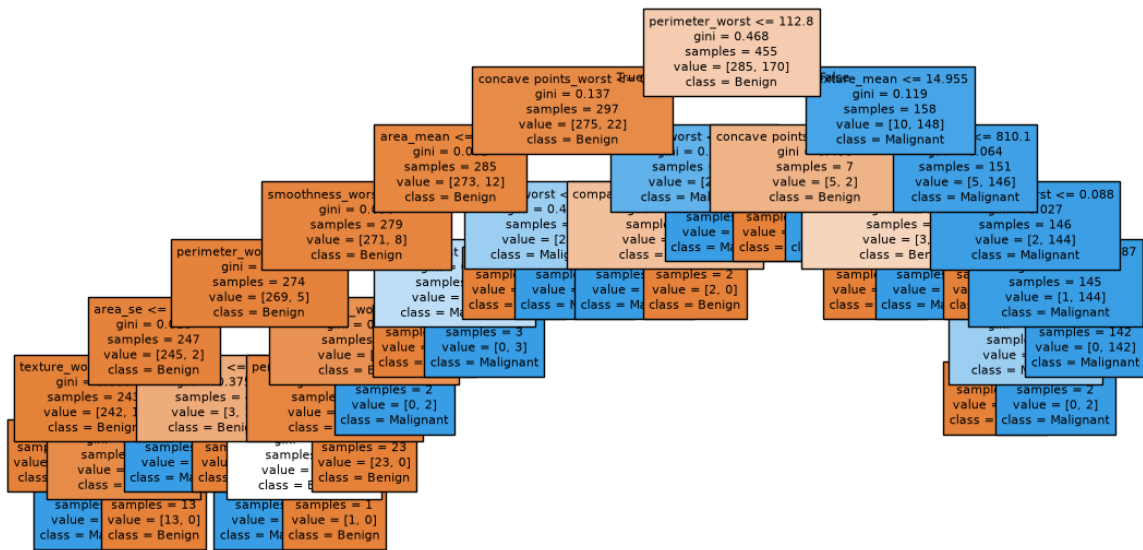
```

    accuracy_score(y_test, rf_pred))
importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": rf.feature_importances_
})
importance = importance.sort_values(
    by="Importance",
    ascending=False
)
print("\nFeature Importances")
print(importance)
plt.figure(figsize=(10,8))
plt.barh(
    importance["Feature"],
    importance["Importance"]
)
plt.gca().invert_yaxis()
plt.xlabel("Importance")
plt.title("Random Forest Feature Importance")
plt.show()
scores = cross_val_score(
    rf,
    X,
    y,
    cv=5
)
print("\nCross Validation Scores:", scores)
print("Average CV Accuracy:", scores.mean())

```

Output:

Decision Tree Accuracy: 0.9298245614035088



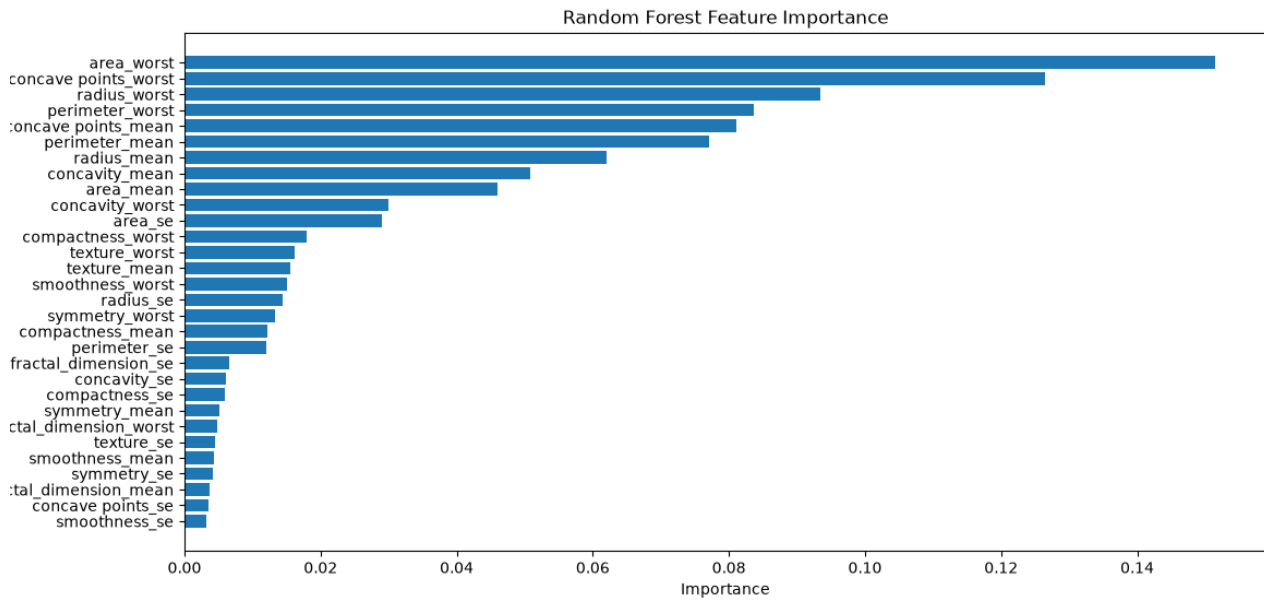
Decision Tree Accuracy (max_depth=3): 0.9035087719298246

Random Forest Accuracy: 0.9736842105263158

Feature Importances

	Feature Importance
23	area_worst 0.151412
27	concave_points_worst 0.126497
20	radius_worst 0.093475
22	perimeter_worst 0.083642
7	concave_points_mean 0.081082
2	perimeter_mean 0.077126
0	radius_mean 0.061990
6	concavity_mean 0.050818
3	area_mean 0.045916
26	concavity_worst 0.030022
13	area_se 0.029064
25	compactness_worst 0.018005
21	texture_worst 0.016174
1	texture_mean 0.015545
24	smoothness_worst 0.015093
10	radius_se 0.014503
28	symmetry_worst 0.013301
5	compactness_mean 0.012218
12	perimeter_se 0.012037
19	fractal_dimension_se 0.006592
16	concavity_se 0.006083
15	compactness_se 0.005929
8	symmetry_mean 0.005097
29	fractal_dimension_worst 0.004887
11	texture_se 0.004436
4	smoothness_mean 0.004349
18	symmetry_se 0.004242
9	fractal_dimension_mean 0.003731

17 concave points_se 0.003544
14 smoothness_se 0.003191



Cross Validation Scores: [0.92105263 0.93859649 0.98245614 0.96491228 0.97345133]

Average CV Accuracy: 0.9560937742586555